

Q1 - 25 July - Shift 2

Let $A : \{1, 2, 3, 4, 5, 6, 7\}$. Define $B = \{T \subseteq A : \text{either } 1 \notin T \text{ or } 2 \in T\}$ and $C = \{T \subseteq A : \text{the sum of all the elements of } T \text{ is a prime number}\}$. Then the number of elements in the set $B \cup C$ is _____

*Space for your notes:***Q2 - 26 July - Shift 2**

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \emptyset\}$ is _____

*Space for your notes:***Q3 - 27 July - Shift 1**

Let R_1 and R_2 be two relations defined on $\mathbb{R}$ by $a R_1 b \Leftrightarrow ab \geq 0$ and $a R_2 b \Leftrightarrow a \geq b$, then

(A) R_1 is an equivalence relation but not R_2

(B) R_2 is an equivalence relation but not R_1

(C) both R_1 and R_2 are equivalence relations

(D) neither R_1 nor R_2 is an equivalence relation

*Space for your notes:***Q4 - 28 July - Shift 1**

Questions

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For $\alpha \in \mathbb{N}$, consider a relation R on $\mathbb{N}$ given by

$R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$. The relation R is an equivalence relation if and only if :

- (A) $\alpha = 14$
- (B) α is a multiple of 4
- (C) 4 is the remainder when α is divided by 10
- (D) 4 is the remainder when α is divided by 7

Space for your notes:

Q5 - 29 July - Shift 1

Let R be a relation from the set $\{1, 2, 3, \dots, 60\}$ to itself such that $R = \{(a, b) : b = pq, \text{ where } p, q \geq 3 \text{ are prime numbers}\}$. Then, the number of elements in R is :

- (A) 600
- (B) 660
- (C) 540
- (D) 720

Space for your notes:

Answer Key**Q1 (107)****Q2 (112)****Q3 (D)****Q4 (D)****Q5 (B)**

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Q1 (107)

(107)

Q2 (112)

 $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$ Total subset of $A = 2^7 = 128$ $C \cap B = \phi$ when set C contains the element

1, 2, 4, 5

 $\therefore S = \{C \subseteq A; C \cap B \neq \phi\}$ $= \text{Total} - (C \cap B = \phi)$ $= 128 - 2^4 = 112$

Q3 (D)

 $R_1 = \{xy \geq 0, x, y \in R\}$ For reflexive $x \times x \geq 0$ which is true.

For symmetric

If $xy \geq 0 \Rightarrow yx \geq 0$ If $x = 2, y = 0$ and $z = -2$ Then $x.y \geq 0$ & $y.z \geq 0$ but $x.z \geq 0$ is not true $\Rightarrow$ not transitive relation. $\Rightarrow R_1$ is not equivalence R_2 if $a \geq b$ it does not implies $b \geq a$ $\Rightarrow R_2$ is not equivalence relation $\Rightarrow D$

Q4 (D)For R to be reflexive $\Rightarrow x R x$

$$\Rightarrow 3x + \alpha x = 7x \Rightarrow (3 + \alpha)x = 7K$$

$$\Rightarrow 3 + \alpha = 7\lambda \Rightarrow \alpha = 7\lambda - 3 = 7N + 4, K, \lambda, N \in \mathbb{I}$$

 $\therefore$ when α divided by 7, remainder is 4.R to be symmetric $xRy \Rightarrow yRx$

$$3x + \alpha y = 7N_1, 3y + \alpha x = 7N_2$$

$$\Rightarrow (3 + \alpha)(x + y) = 7(N_1 + N_2) = 7N_3$$

Which holds when $3 + \alpha$ is multiple of 7 $\therefore \alpha = 7N + 4$ (as did earlier)

R to be transitive

$$xRy \text{ \& } yRz \Rightarrow xRz$$

$$3x + \alpha y = 7N_1 \text{ \& } 3y + \alpha z = 7N_2 \quad \text{and}$$

$$3x + \alpha z = 7N_3$$

$$\therefore 3x + 7N_2 - 3y = 7N_3$$

$$\therefore 7N_1 - \alpha y + 7N_2 - 3y = 7N_3$$

$$\therefore 7(N_1 + N_2) - (3 + \alpha)y = 7N_3$$

$$\therefore (3 + \alpha)y = 7N$$

Which is true again when $3 + \alpha$ divisible by 7, i.e.when α divided by 7, remainder is 4.**Q5 (B)**

Number of possible values of $a = 60$, for $b = pq$,

If $p = 3, q = 3, 5, 7, 11, 13, 17, 19$

If $p = 5 \quad q = 5, 7, 11$

If $p = 7 \quad q = 7$

Total cases $= 60 \times 11 = 660$